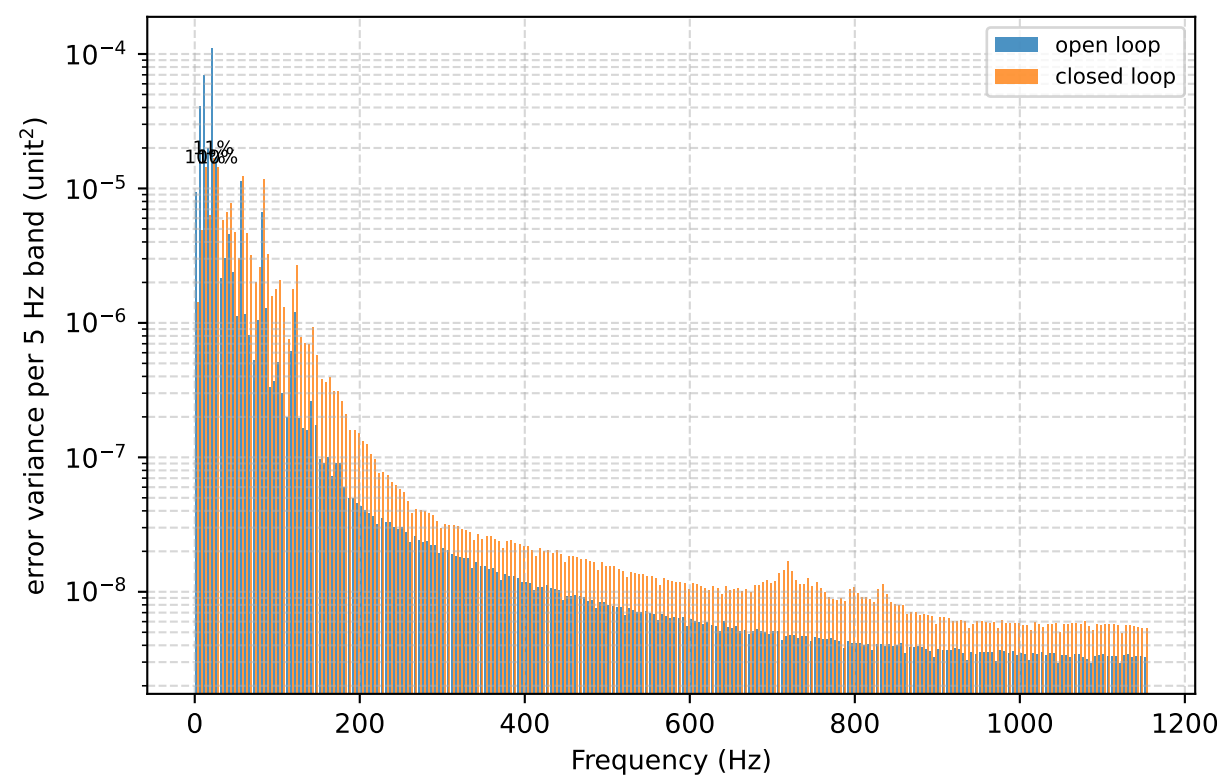
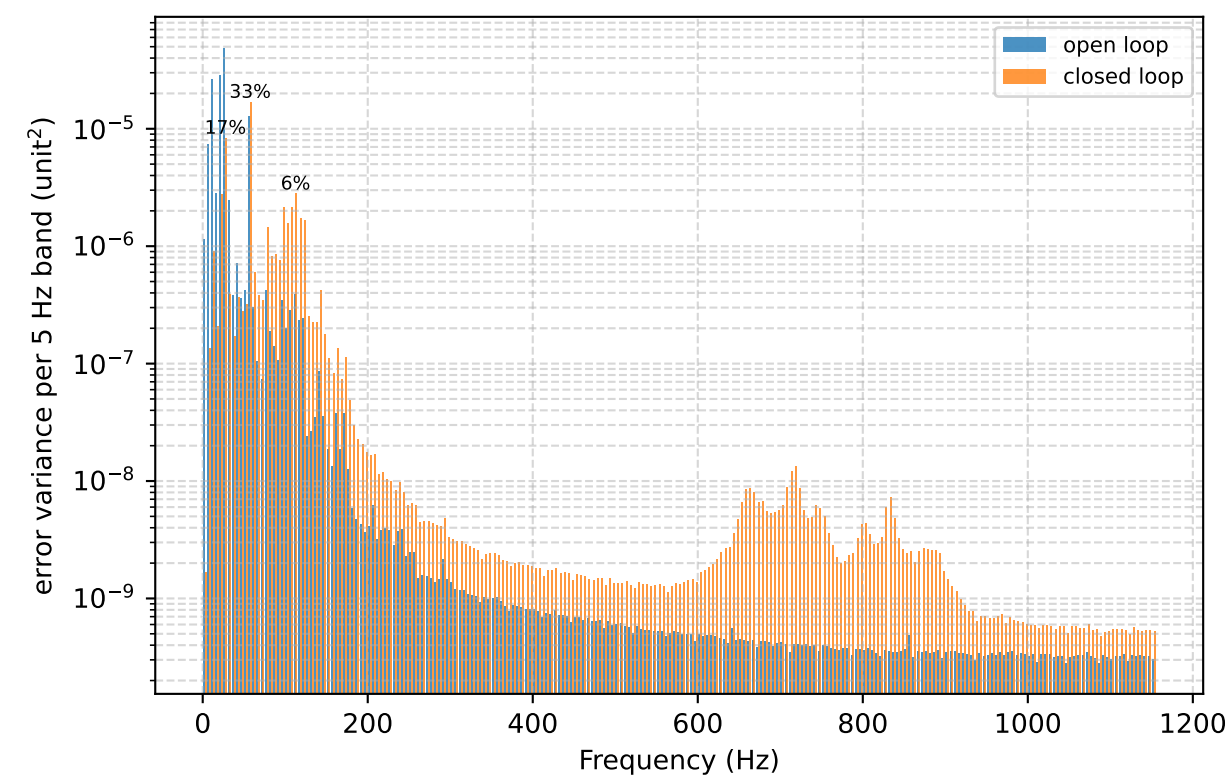
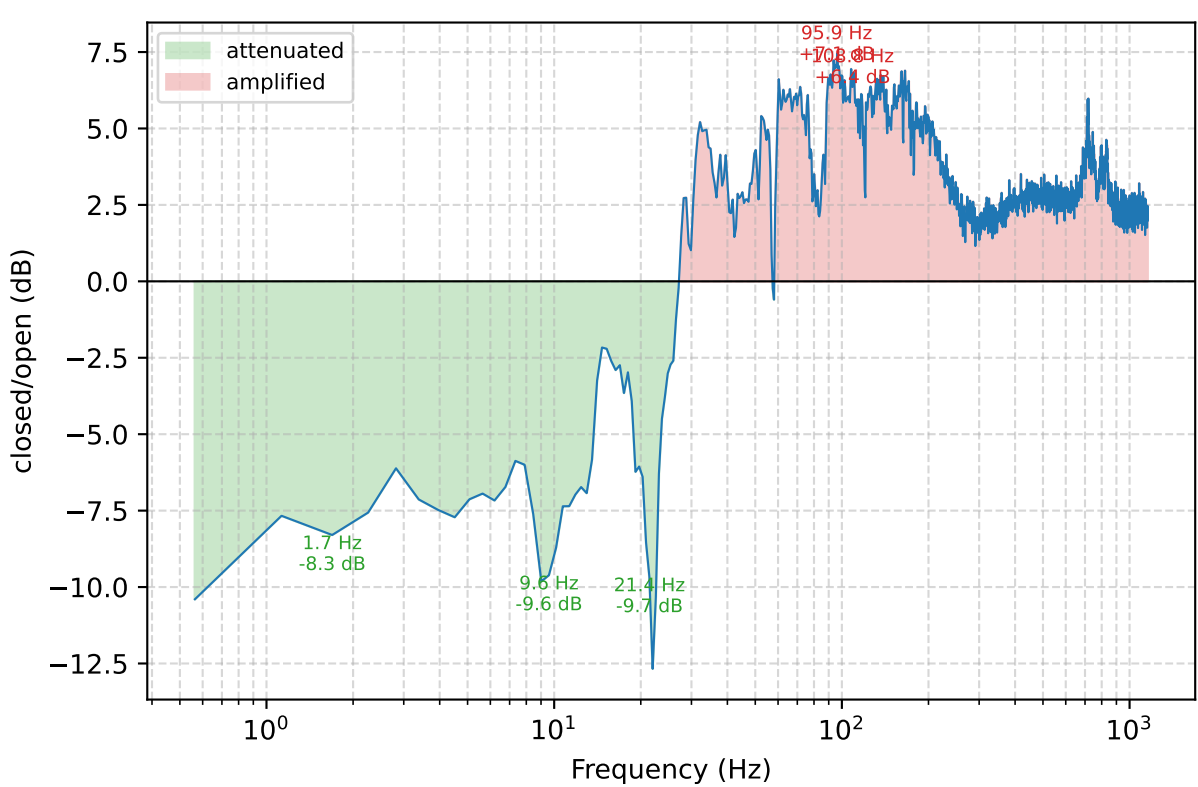
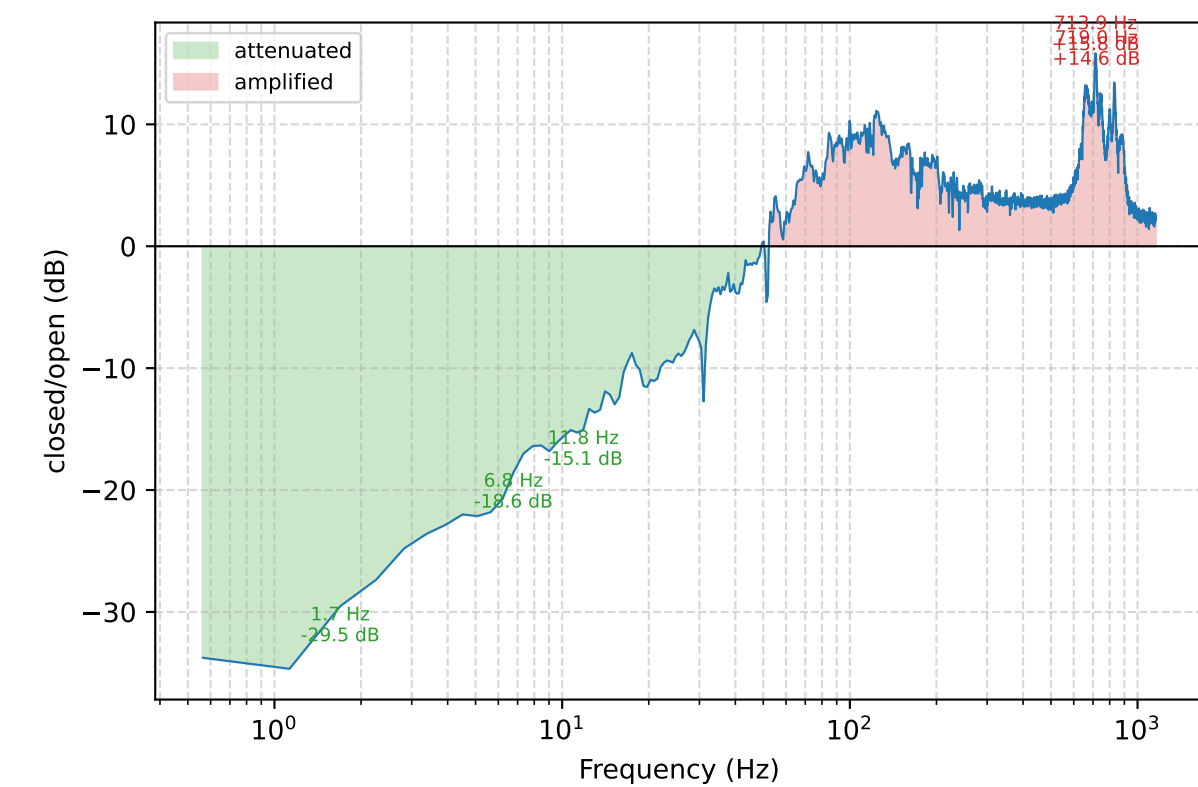
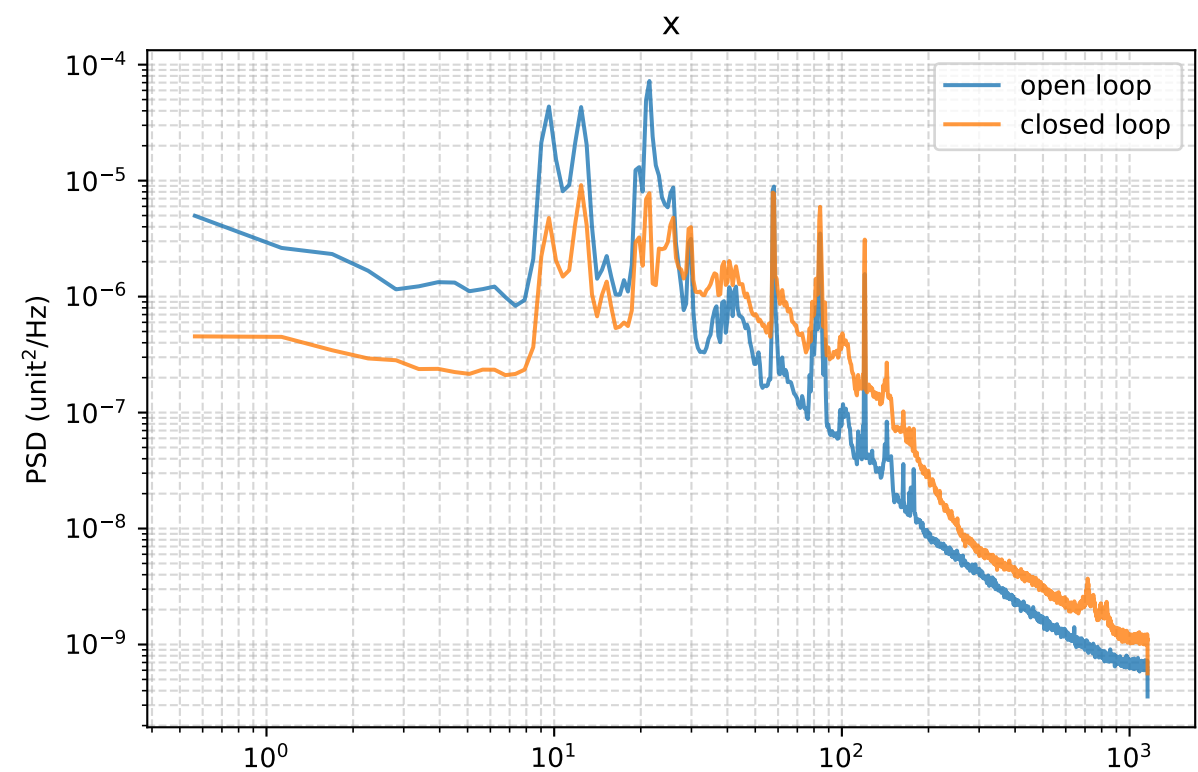
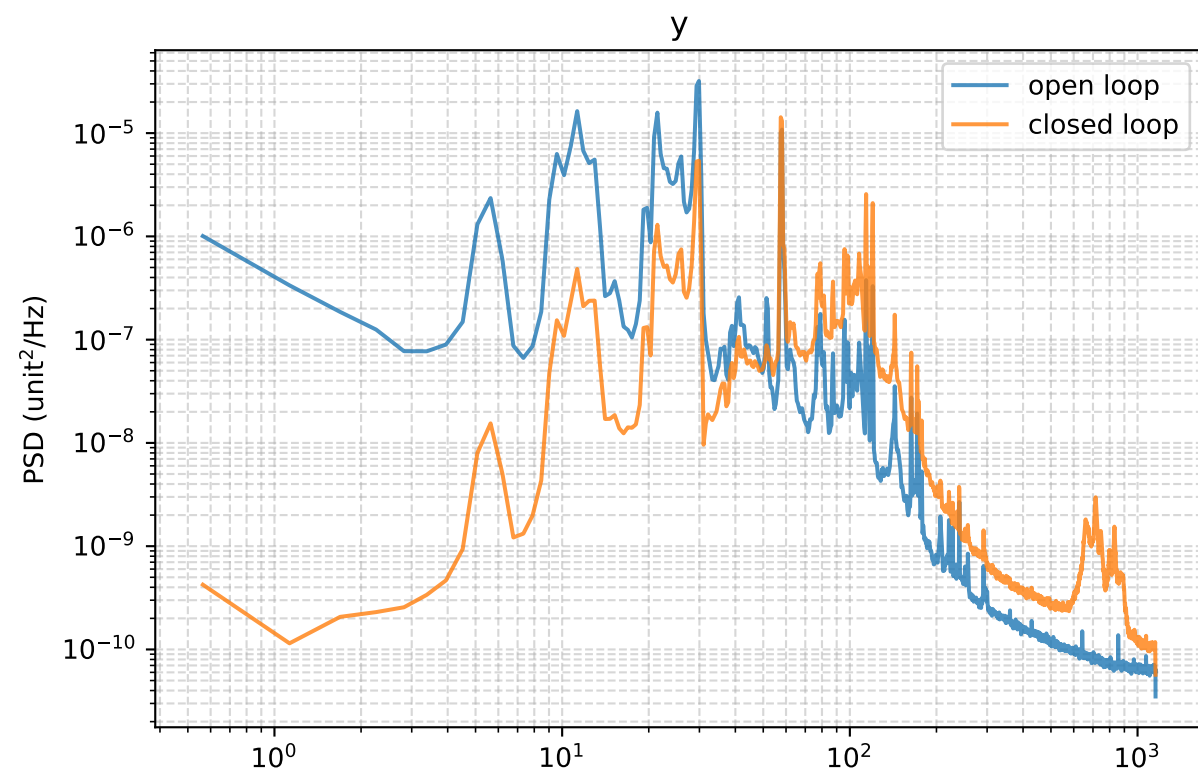


Closed-loop attenuation



Run configuration and per-band RMS

Started: 2026-07-13 12:52:31
pixel_scale: 31.5 px per output unit
y: mean -4.858 -> +0.000 px; RMS about mean 0.403 -> 0.225 px (open -> closed)
x: mean +5.451 -> -0.003 px; RMS about mean 0.635 -> 0.384 px (open -> closed)

Config used:
100, 500, 100, 500, 4096: camera ASI290MM 64x64, ROI start (920,488), ~2310 Hz loop
pid: integrator-only (p=py=d=0), i=600 iy=1300, units=minmax
lines x=[22.08,57.3] gain=[60,20] phase=[56,3]
linesy=[30.8] gainy=[60] phasey=[60], line_delay=0.002
kalman: order=60 horizon=5 mu=0.03 gain=1.0 gainy=0 ramp=10
piplate: fine ch0(+2)/ch1(-2), coarse parked, baud 460800, start_delay: 10.0, open_time: 500.0, settle_time: 40.0, closed_time: 500.0

RMS contributed by each 10 Hz band (px)				
band (Hz)	y open	y closed	x open	x closed
0-10	0.0925	0.0116	0.2243	0.0792
10-20	0.1705	0.0333	0.2993	0.1437
20-30	0.2782	0.1054	0.3572	0.1759
30-40	0.0532	0.0237	0.0717	0.1112
40-50	0.0328	0.0255	0.0829	0.1114
50-60	0.1147	0.1312	0.1111	0.1236
60-70	0.0201	0.0313	0.0443	0.0883
70-80	0.0221	0.0422	0.0397	0.0676
80-90	0.0180	0.0408	0.0891	0.1224
90-100	0.0213	0.0538	0.0264	0.0578
100-110	0.0219	0.0607	0.0284	0.0581
110-120	0.0248	0.0671	0.0285	0.0501
120-130	0.0163	0.0437	0.0372	0.0585
130-140	0.0078	0.0212	0.0180	0.0373
140-150	0.0110	0.0244	0.0208	0.0387
150-160	0.0056	0.0139	0.0137	0.0272
160-170	0.0075	0.0145	0.0131	0.0264
170-180	0.0071	0.0126	0.0134	0.0239
180-190	0.0033	0.0072	0.0104	0.0191
190-200	0.0028	0.0062	0.0098	0.0176
200-210	0.0032	0.0058	0.0091	0.0160
210-220	0.0026	0.0048	0.0086	0.0142
220-230	0.0028	0.0045	0.0082	0.0124
230-240	0.0026	0.0042	0.0081	0.0117
240-250	0.0025	0.0038	0.0077	0.0109
250-260	0.0022	0.0036	0.0076	0.0101
260-270	0.0017	0.0030	0.0070	0.0089
270-280	0.0017	0.0030	0.0069	0.0089
280-290	0.0017	0.0029	0.0068	0.0087
290-300	0.0019	0.0028	0.0065	0.0079
total 0-300	0.3681	0.2237	0.5559	0.3805